

Python Basics — RAG Practice Notes

1. What is Python?

Python is a high-level, interpreted programming language known for its simple and readable syntax. It is commonly used for web development, automation, data science, machine learning, and artificial intelligence.

2. Variables

A variable is used to store data.

```
name = "Shashank"  
age = 22
```

3. Data Types

```
Common Python data types include:  
int – whole numbers  
float – decimal numbers  
str – text  
bool – True or False  
list – ordered collection  
tuple – immutable collection  
dict – key-value pairs  
set – collection of unique values
```

4. Conditional Statements

Conditional statements allow a program to make decisions.

```
age = 20  
  
if age >= 18:  
    print("Adult")  
else:  
    print("Minor")
```

5. Loops

A for loop repeats code for each item in a sequence.

```
for i in range(5):  
    print(i)
```

A while loop continues while a condition is true.

```
count = 0  
while count < 5:  
    print(count)  
    count += 1
```

6. Functions

A function is a reusable block of code.

```
def add(a, b):  
    return a + b  
  
result = add(10, 20)  
print(result)
```

7. Lists

A list stores multiple values.

```
numbers = [10, 20, 30, 40]
```

```
print(numbers[0])
```

8. Dictionaries

A dictionary stores data as key-value pairs.

```
student = {  
    "name": "Shashank",  
    "age": 22,  
    "course": "Computer Science"  
}
```

```
print(student["name"])
```

9. Object-Oriented Programming

Python supports object-oriented programming using classes and objects.

```
class Student:  
    def __init__(self, name):  
        self.name = name  
  
student = Student("Shashank")  
print(student.name)
```

10. Exception Handling

Exceptions can be handled using try and except.

```
try:  
    result = 10 / 0  
except ZeroDivisionError:  
    print("Cannot divide by zero")
```